

Unit 3, Lesson 3: Multiplying Rational Numbers

Benchmark

MA.7.NSO.2.2 Add, subtract, multiply, and divide rational numbers with procedural fluency.

Learning Target

- ☐ I can fluently multiply rational numbers.

3.3.1 Warm-Up: Bell Work

1. Using the digits 1 to 6 at most one time each, fill in the boxes so that the top two equations are equal, and the bottom equation has the greatest value.

$$- \square + \square =$$

$$- \square - \square =$$

$$- \square - \square =$$

3.3.2 Collaboration: Signed Number Rules

1. Fill in the blank with the words *positive* or *negative* to describe the rules for multiplying integers.
- The product of two positive integers is a _____ integer.

The product of two negative integers is a _____ integer.

The product of a negative integer and a positive integer is a _____ integer.
2. Discuss with your partner whether you think the same rules apply to rational numbers. Write a summary of your reasoning.
3. Complete the table. Use a calculator to find the product of each expression to verify if the same rules for signs apply when multiplying rational numbers.

Expression	Sign of Product Using the Rules	Product	Did the Same Rule Apply?
$\left(-\frac{3}{5}\right)\left(-\frac{3}{11}\right)$			
$\frac{7}{18}\left(-4\frac{8}{25}\right)$			
$21.285 \cdot -3.8$			
$(-91.09)(-6.4)$			

4. Complete the sentence.
- The rules for multiplying rational numbers are

the same as

different from

the integer

rules for multiplication.

3.3.3 Guided Practice: Applying Signed Number Rules to Rational Numbers

1. With your partner, decide who will be Partner A and who will be Partner B. Then, find the product of the expression in the column that matches your letter.

Column A	Product	Column B
$790\left(\frac{1}{10}\right)$		$7.9(10)$
$\left(-\frac{6}{7}\right)\cdot 7$		$(0.1)\cdot (-60)$
$2.1\cdot (-2)$		$(-8.4)\left(\frac{1}{2}\right)$
$2.5\cdot -3.25$		$\left(-\frac{5}{2}\right)\left(\frac{13}{4}\right)$

The expressions in each row have the same product. Check your work with your partner and, if necessary, resolve any differences. Write the agreed-upon product in the center column.

2. Choose a value from Row A and a value from Row B that, when multiplied, will result in a negative answer. Then, calculate the product.

Row A

$-9\frac{2}{7}$	$-\frac{3}{5}$	$12\frac{2}{9}$
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Row B

$4\frac{3}{8}$	$\frac{11}{16}$	$-\frac{9}{50}$
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Row A

×

Row B

=

Product

3.3.4 Wrap-Up: Ticket Out the Door

1. Find the product of each expression.

a. $(0.24)(-3.8)$

b. $\left(-\frac{2}{5}\right)\left(-\frac{9}{8}\right)$

2. Fill in the blanks with the words *positive* or *negative*.

- Two positive factors result in a _____ product.
- Two negative factors result in a _____ product.
- A positive factor and a negative factor result in a _____ product.

